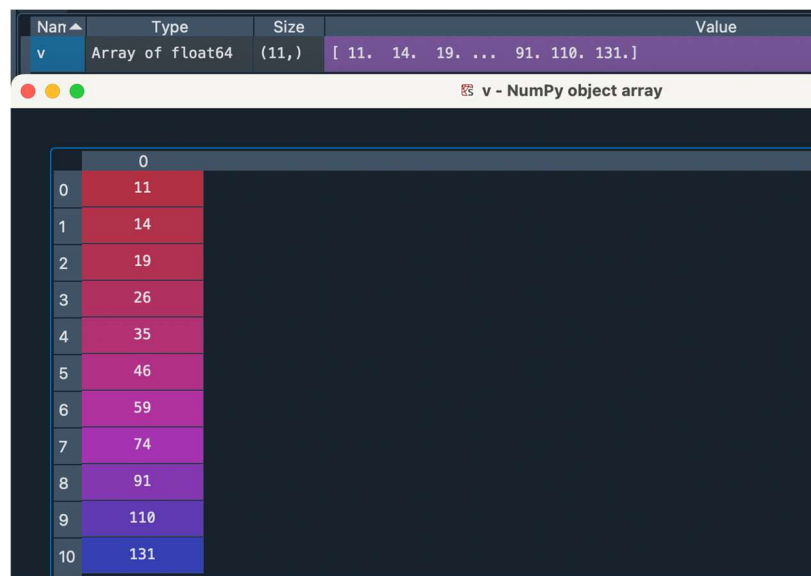


- Indexing arrays and matrices is an important aspect of programming syntax to understand.
 - Note indexing follows the row x column standard. For example given a 2-d matrix A: A[row index, column index]. **Be careful as Python indexing starts at zero!**
 - If you need to select more than 1 index, use colon ":" to specify an array of indices. For example b[0:3] would select the first 3 values in the b array. **Note: the Python ending index stops before the value you want (i.e. 3 here, so Python will return b[0], b[1], and b[2] with this syntax) (which is really annoying) so you basically have to add 1 to your final index if you are selecting multiple elements in an array.**
 - If only a colon ":" is used (with no beginning and end indices) to index a dimension of an array, all indexes of that array are selected. For example A[:,0] would select all rows and the first column of matrix A.
 - Lastly, to index an array or matrix to the end of a dimension minus some number of indices you can use the indexing syntax [-1]. For example b[:-1] selects all values of array b except for the last index. Or b[1:-1] selects all values of array b except for the first and last indexes.

Complete the following indexing exercises:

Given the Python variable window shown below:



	0
0	11
1	14
2	19
3	26
4	35
5	46
6	59
7	74
8	91
9	110
10	131

- Index the 10th element of v
v[9]
- Index the 4th through 7th elements of v
v[3:6]
- What values are returned if question b. was entered into the command window or run in a script?
[26, 35, 46, 59]

Given the Python variable window shown below:

Name	Type	Size	
A	Array of float64	(4, 4)	[[4. 19. 7. 11.] [1. 8. 3. 15.]

A - NumPy object array				
	0	1	2	3
0	4	19	7	11
1	1	8	3	15
2	1	0	0	7
3	6	17	4	13

- d. Index the value of A in the 3rd row and 2nd column.
A[2,1]
- e. What value is returned if question d. was entered into the command window or run in a script?
0
- f. Index A to select the 2nd row and all columns of A.
A[1,:]
- g. What values are returned if question f. was entered into the command window or run in a script?
[1, 8, 3, 15]
- h. Index A to select the 4th column and all rows except for the last row.
A[:-1, 4]
- i. What values are returned if question h. was entered into the command window or run in a script?
[11, 15, 7]
- j. Lets say T is a 5 row x 50 column matrix preallocated with zeros (a must when creating and working with arrays and matrices with for loops in Python). This line of code for preallocation would read:

```
import numpy as np
T=np.zeros([5,50])
```

Use two for loops to index every element of the matrix and set that element equal to $i*j^2$ by the time the for loops are completed. I have provided the correct double for loop syntax below. Fill in the correct indexing inside the for loops:

```
for i in range(T.shape[1]): # iterate through the columns
    for j in range(T.shape[0]): # iterate through the rows
        T[i,j] = i*j^2;
```

- k. You wish to calculate the average mass from 2 sets of measurements given in the first and second rows of variable m shown below:

	0	1	2	3	4	5	6	7	8	9
0	5	9	2	1	1	6	8	7	3	4
1	4	7	1	2	3	5	9	8	3	5

While a mean function does exist in Python to accomplish this, please write 1 line of code manually (for indexing practice) in the given for loop to calculate the average mass for each trial in a variable called mavg:

```
import numpy as np  
mavg=np.zeros(m.shape[1])
```

```
for i in range(m.shape[1]):
```

```
    mavg[i] = (m[0,i] + m[1,i])/2;
```

- I. Write the code to calculate the average mass from part k using matrix notation instead (much faster for Python to compute when dealing with large arrays and matrices).

```
mavg = np.mean(m, axis=0);
```

2. A truss is loaded as shown in Figure 2.

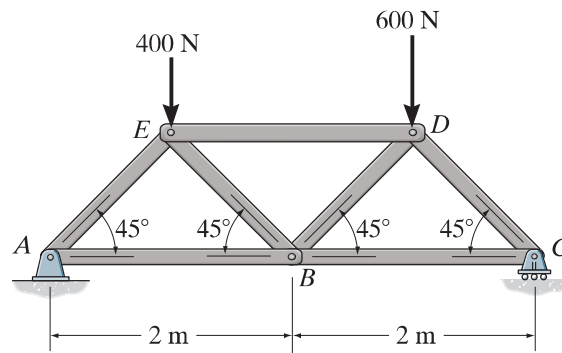


Figure 2: Truss loading

You are given the following sets of equations based on Method of Joints for joints A, B, and C.

$$A_x + AB + AE \cos(45) = 0$$

$$A_y + AE \sin(45) = 0$$

$$-AB + BC - BE \cos(45) + BD \cos(45) = 0$$

$$BE \sin(45) + BD \sin(45) = 0$$

$$-BC - CD \cos(45) = 0$$

$$C_y + CD \sin(45) = 0$$

Assuming all members are in tension for sign consistency, write four additional equations using Method of Joints at joint D and E.

E: $+ED - EA \cos(45) + EB \cos(45) = 0$ $-400 - EA \sin(45) - EB \sin(45) = 0$	D: $-ED - DB \cos(45) + DC \cos(45) = 0$ $-600 - DB \sin(45) - DC \sin(45) = 0$
--	--

Solve for the 10 unknowns, AB , AE , BC , BD , BE , CD , DE , A_x , A_y , and C_y . For the sake of consistency (and my sanity for grading), let's all assume the same organization of our unknown vector:

$$x = \begin{Bmatrix} AB \\ AE \\ BC \\ BD \\ BE \\ CD \\ DE \\ A_x \\ A_y \\ C_y \end{Bmatrix}$$

[450., -636.39610307, 550., -70.71067812, 70.71067812, -777.81745931, -500., 0., 450., 550.]

3. A wall is constructed of concrete (grey) and insulation (blue) as shown in Figure 1. At equilibrium, the temperature profile (T_1 , T_2 , and T_3) in the wall is determined given T_0 and T_4 ambient air temperatures inside and outside the wall respectively. Heat transfer to the surfaces of the wall are governed by Newton's Law of Cooling: $q = h (T_f - T_i)$. Heat transfer through the wall materials are governed by Fourier's Law: $q = k/L (T_f - T_i)$.

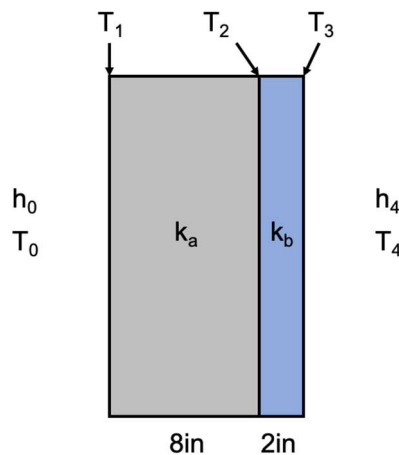


Figure 1: Composite slab subject to temperature boundary conditions

Write the 4 equations for heat transfer from the inside air into the concrete (q_1), through the concrete (q_2), through the insulation (q_3), and into the outside air (q_4):

To solve the system of equations, each heat transfer must equal according to conservation of energy $q_1 = q_2$, $q_2 = q_3$, and $q_3 = q_4$. Express these 3 equations in matrix form: